

LatentVerse: A Framework for Understanding Shared and Modality-Specific Information in Multimodal Latent Representations

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Abstract

Latent embeddings have become a central data abstraction in modern machine learning, especially in biomedicine, where foundation models are increasingly used to encode multimodal data like clinical text, medical images, omics, and physiological signals. However, the utility and value of these representations depends on understanding their quality, structure, and the information they encode. Existing analysis workflows for evaluating representations remain fragmented across custom scripts, isolated metrics, and most importantly lack multimodal analysis, limiting accessibility and reproducibility. We present LatentVerse, a representation analysis resource that combines a web-based visual analytics platform for accessible, report-driven exploration with a command-line interface for scalable technical workflows. LatentVerse unifies diagnostics for various representation quality metrics and extends to multimodal settings by decomposing embeddings into shared and modality-specific components. We evaluate LatentVerse through controlled unimodal and multimodal simulations, discovery-oriented analyses on real biomedical embeddings, and a user study across diverse use cases. By supporting thorough and interpretable evaluation of latent spaces, LatentVerse makes foundation model representations more understandable in biomedical and data science applications.

1 Introduction

Latent representations (embeddings) have become a standard data abstraction in modern machine learning, in applications ranging from natural language to vision [3]. In particular, the recent shift toward large-scale pretraining followed by task-specific adaptation has further increased their importance [1, 4]. Embeddings enable knowledge discovery [14], are reused across tasks [2, 21], compared across models and checkpoints in zero-shot settings [15], and shared across teams or institutions when raw data cannot be easily accessed [11]. As a result, understanding the quality of an embedding space has become an important part of modern machine learning workflows.

This need is especially pressing when downstream evaluation alone is insufficient [33]. Practitioners often need to compare representations before expensive fine-tuning and assess whether embeddings remain reliable under limited labels, noisy inputs, or domain shift. These challenges are even greater in multimodal settings, which are becoming increasingly common across real-world machine learning applications. Unlike unimodal embeddings, multimodal representations may entangle information from various sources, yet there are few established tools for systematically evaluating this structure. Representation-centric diagnostics, therefore, provide a more direct way to evaluate latent spaces, using metrics such as clusterability, disentanglement, robustness, expressiveness, and predictability to characterize how information is organized, preserved, and exposed within both unimodal and multimodal embeddings.

Recent scientific machine learning resources have highlighted the growing value of accessible and reproducible analysis platforms, yet existing tools only partially address representation evaluation. General-purpose embedding viewers such as the TensorBoard Embedding Projector [25], Emblaze [24], WizMap [31], and Embedding Atlas [22] support interactive projection and neighborhood exploration, while domain-specific systems such as Latent Space Explorer and other scientific web applications provide guided analysis within particular modalities or application areas [6, 17, 19]. In parallel, benchmarking and data platforms emphasize downstream task performance or large-scale data access rather than representation-centric diagnosis [7, 9, 13]. Taken together, these tools establish the value of interactive exploration and accessible infrastructure, but none provides a unified, model-agnostic workflow for systematic latent-space evaluation that spans unimodal and multimodal settings and can localize where information resides across modalities.

To address this gap, we present LatentVerse, a representation-analysis tool that builds on the original LatentVerse toolkit [29]. LatentVerse integrates a broad family of representation diagnostics into a unified, model-agnostic framework and extends it to multimodal representations through information decoupling [28, 30], which separates shared from modality-specific latent information so that each component can be analyzed independently. This makes it possible to reason directly about where useful information resides in multimodal systems and how different representation components contribute to performance and interpretability.

We evaluate LatentVerse along three axes: controlled unimodal and multimodal simulations that test whether its diagnostics recover known structure; real biomedical embedding analyses that show its utility for discovery; and a user study that assesses its practical value in analytical workflows. By integrating these capabilities into a single workflow, LatentVerse helps bridge the gap between representation learning research and applied practice.

2 The LatentVerse Framework

LatentVerse is a model-agnostic framework composed of a web application and a command-line interface for evaluating latent representations. Users provide one or more representation matrices and optional labels, select a diagnostic test, and receive a structured report combining quantitative metrics, interactive visualizations, and a descriptive summary. The framework supports both unimodal evaluation and multimodal analysis described in Section 2.4. Figure 1 provides an overview of the system and user workflow. The high-level framework is described in the following sections, and more detail on implementation, deployment, and interface engineering is presented in Appendix A.1.

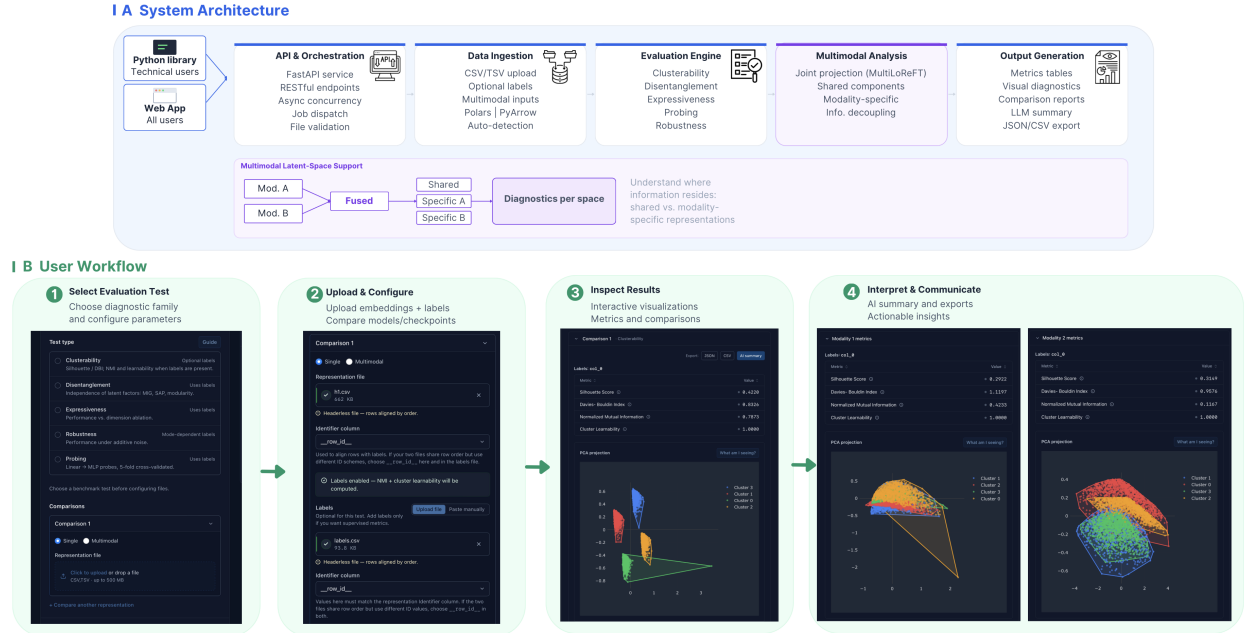


Figure 1: **Overview of the LatentVerse framework.** A) System architecture showing the shared evaluation pipeline, from web and Python-library access through data ingestion, representation diagnostics, multimodal decomposition, and report generation. B) User workflow, from selecting an evaluation test and uploading embeddings to inspecting results and communicating findings through exportable reports and AI-generated summaries.

2.1 Representation quality dimensions

LatentVerse evaluates five complementary properties of latent representations: clusterability, disentanglement, expressiveness, downstream predictability, and robustness. Together, these provide a multi-faceted view of representation quality that goes beyond downstream task accuracy alone. LatentVerse uses established metrics and tests for each evaluation category to provide a consistent workflow. Detailed description and definitions of metrics are provided in Appendix A.3.

Clusterability. The clusterability test evaluates whether a representation forms compact and well-separated groups in latent space. LatentVerse combines intrinsic metrics, including Silhouette Score [23] and Davies–Bouldin Index [10], with label-aware measures such as Normalized Mutual Information (NMI) [26] and Cluster Learnability [18]. The resulting report helps users determine whether the embedding supports meaningful grouping structure and whether that structure aligns with known labels.

Disentanglement. The disentanglement test evaluates whether distinct factors of variation are encoded in separate and interpretable latent dimensions. LatentVerse reports complementary metrics including the DCI framework [12], Mutual Information Gap (MIG) [8], Separated Attribute Predictability (SAP) [16], and Total Correlation (TC) [32]. These outputs help users distinguish representations that are cleanly factorized from those in which relevant information is spread across many correlated dimensions.

Expressiveness. The expressiveness test evaluates how efficiently predictive information is distributed across the latent dimensions. To measure this, LatentVerse performs dimension ablation by progressively removing the highest-variance dimensions and measuring the resulting drop in predictive performance, summarizing this behavior through Compactness and Intrinsic Dimension. This lets users assess whether useful signal is concentrated, redundant, or diffusely distributed.

Downstream predictability. The downstream predictability test measures how much task-relevant information can be recovered from the latent space. LatentVerse evaluates a family of supervised probes of increasing complexity, from linear models to multilayer perceptrons, and reports standard predictive metrics such as Accuracy, AUROC, macro-F1, and R^2 , depending on the task. This distinguishes representations that are directly predictive, those that require more complex decoders, and those that contain little usable signal for the target label.

Robustness. The robustness test evaluates how stable a representation remains under perturbations. LatentVerse injects Gaussian noise into the latent space and tracks the resulting degradation in clustering- or probing-based performance. The corresponding degradation curves reveal whether an embedding is resilient or fragile under noise, providing a practical view of representation reliability.

2.2 A unified evaluation workflow

The framework is organized around a three-step workflow. First, the user selects the type of representation they want to evaluate (unimodal or multimodal) and an evaluation test along with the test-specific options. Second, they upload one or more representation files together with optional labels and comparison settings. Third, LatentVerse executes the requested analyses and renders the results as a report consisting of an interactive visualization, a metrics table, exportable outputs, and an optional descriptive summary. Crucially, the same pipeline drives unimodal and multimodal analysis and is exposed identically through the

web application and a command-line interface, so that analyses can move between accessible exploration and scalable, scriptable workflows without changing the underlying diagnostics. A parity test suite verifies that the two interfaces produce numerically identical results—agreement to a relative tolerance of 10^{-9} on the same inputs—so an analysis can be developed interactively and rerun from a script without change. Implementation details that support practical use on large embedding matrices, scalable ingestion, asynchronous job execution, and resumable uploads, are described in Appendix A.1.

2.3 Visual analytics and report generation

The interface is designed to support both technical and less technical users. Guided configuration, built-in documentation, example reports, and contextual explanations lower the barrier to entry, while comparison cards, advanced options, and export functionality support more detailed analytical workflows. Each report couples three coordinated views, a metrics table, a test-specific interactive visualization, and a on-click natural-language summary, so that numerical evidence, visual evidence, and interpretation are presented together rather than as isolated artifacts. Across all tests, the goal is to make the properties of a latent space easier to inspect, compare, and interpret. The summary is produced by a server-side, template-constrained query to a large language model; details of the visualization types and the summary mechanism are given in Appendix A.2.

2.4 Multimodal analysis

The most notable capability of LatentVerse is its treatment of multimodal representations. Modern machine learning systems increasingly integrate multimodal sources like text, images, audio, or clinical measurements, and the resulting embeddings can be more informative than unimodal ones. They are also substantially harder to interpret because a single fused vector may entangle information that is shared across modalities with information that is specific to each modality.

Prior work in factorized multimodal representation learning has proposed information decoupling as a way to improve interpretability by separating shared factors that capture cross-modal agreement from modality-specific factors that retain unique information from each input stream [27, 30]. LatentVerse integrated this capability into its workflow to separate the shared and modality-specific information in multimodal representation and applies the same family of diagnostics used in the unimodal case not only to a fused multimodal embedding, but also to decomposed components.

To obtain this decomposition, LatentVerse leverages the MultiLoReFT framework [28], a lightweight fine-tuning approach for separating shared and private information in multimodal representations. MultiLoReFT isolates information unique to each modality from information shared across modalities which enables representation evaluation on each component independently. The details of our experiments are reported in Appendix A.4, Table 1.

Decomposition changes the kinds of questions that can be asked of a multimodal representation. Rather than asking only *how predictive is this embedding?*, a user can ask whether a clinically relevant factor is encoded in the shared space or in one modality’s private space; whether predictive structure is genuinely cross-modal or driven by a single modality; and

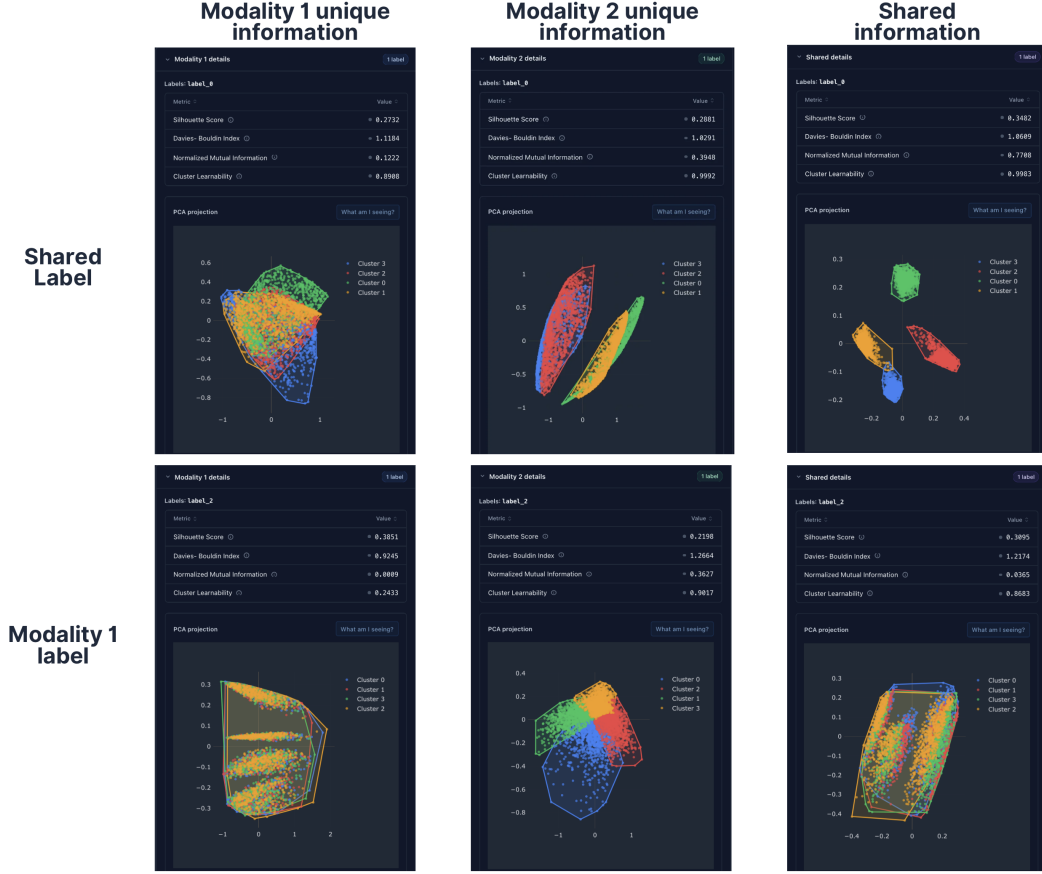


Figure 2: **LatentVerse multimodal analysis.** LatentVerse decomposes information from multiple representations into shared and modality-specific components. By evaluating each component separately, the platform helps users identify where shared and modality-specific information resides in latent space.

whether shared and private components differ in compactness or robustness. By evaluating each component separately, LatentVerse makes these questions answerable from the data rather than assumed.

Figure 2 shows how LatentVerse decomposes a multimodal representation into shared and modality-specific components and how the behavior of each component reveal variability in the underlying generative factors. In this way, LatentVerse moves beyond treating a multimodal embedding as a single fused object and instead turns it into an analyzable representational object. This also makes the framework especially useful where researchers need to compare fusion strategies, validate whether modalities contribute complementary information, or communicate why one multimodal model is preferable to another.

3 Validation on Controlled Data

We first validate LatentVerse on simulated representations with well-defined and known structure. The purpose of these experiments is to confirm that the framework’s integrated

metrics and visual reports recover the intended properties of a latent space, so that further analysis with the goal of scientific discovery (Section 4) rest on trusted diagnostics. Full details of all unimodal and multimodal dataset generation are provided in Appendix A.5.

3.1 Unimodal simulations

We generated dataset families that each target one representation property (clusterability, predictability, disentanglement, and expressiveness), with three variable levels per family ranging from settings where the target property is strongly present to settings where it is weak or absent. Across these datasets, we show that LatentVerse provides evaluation reports that track the known structure, and helps identify such characteristics of the data. As a representative example, the clusterability datasets consists of representations that vary from clearly separated clusters to overlapping groups to no class-aligned structure. Figure 3 shows LatentVerse reports for these representation groups. In the highly clusterable setting the Silhouette Score (0.8936) and NMI (1.0000) reflect clearly separated clusters, and Cluster Learnability (1.0000) confirms the class labels are almost perfectly recoverable from the representation; in the unstructured setting these fall to 0.0103, 0.0007, and 0.2280 respectively—the last near the 0.20 chance level for five classes—with the visual report showing the same progression from tight, well-separated groups to diffuse, overlapping point clouds. LatentVerse report also includes a descriptive summary report that analyses the evaluated metrics and provides an interpretable description of the representation behaviour according to the evaluation category. The predictability, disentanglement, and expressiveness families behave similarly, distinguishing directly accessible from decoder-dependent and uninformative labels, factorized from entangled representations, and concentrated from redundant or sparse information distributions, respectively. Because these results serve only to confirm expected metric behavior, we report the full set of examples and figures in Appendix A.6 and summarize them here.

3.2 Multimodal simulations

We also evaluate LatentVerse in a multimodal setting designed to test whether the framework can provide better insight into the multimodal datasets by decoupling shared from modality-specific information. For this purpose, we generate a multimodal simulated dataset where the representations are generated from independent latent factors drawn from diverse distributions, with labels defined to be shared across modalities, specific to one modality, or conditionally dependent on interactions between modalities (See Appendix A.5.2 for more detail on this dataset). The experiment is intended to show that decomposition yields a more informative view than evaluating only a fused embedding. After decomposing the embedding into shared and modality-specific components, the reports reveal that shared labels are concentrated in the shared latent space, while modality-specific labels are most strongly recovered in their corresponding private components (Figure 2). This separation is reflected in both the reported metrics and the latent-space visualizations, which makes it possible to inspect where different forms of information reside. These controlled results establish that the decomposition workflow recovers the ground-truth organization of information; this motivates its use on real embeddings where no such ground truth is available.

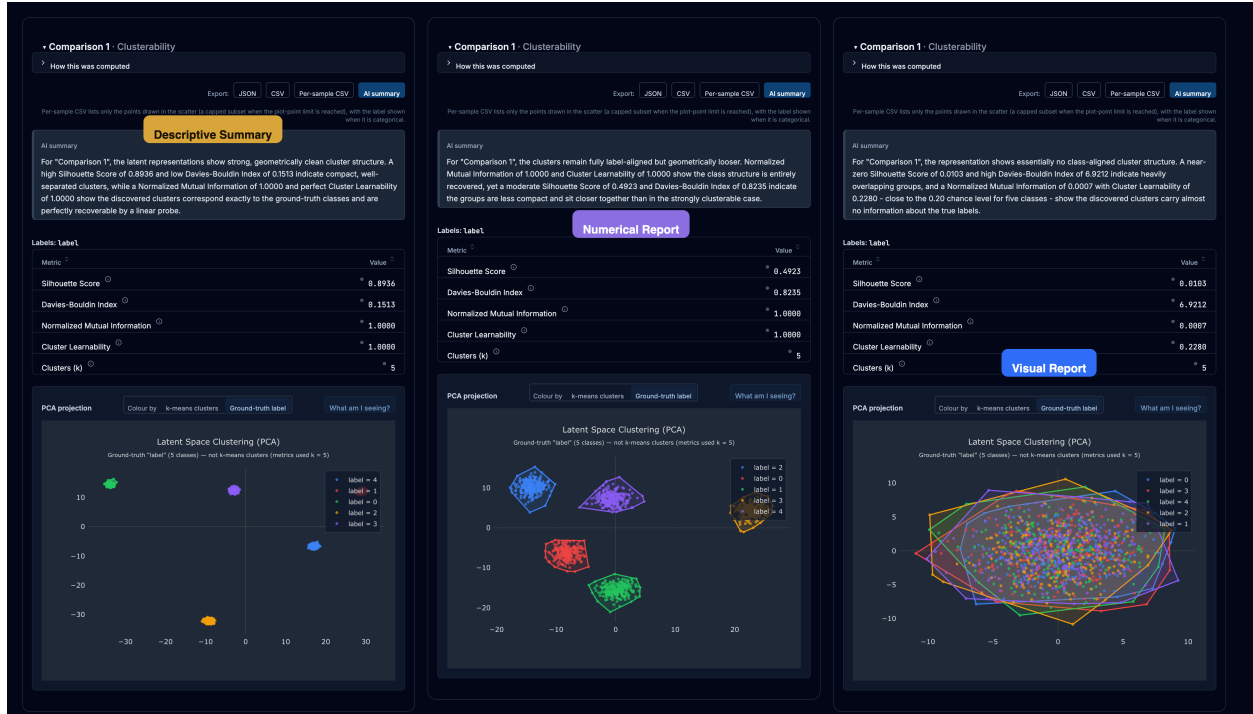


Figure 3: Validation on controlled simulated data. Example LatentVerse reports for representations with varying clusterability. As cluster structure degrades from left to right, the reported metrics reflect weaker class-aligned grouping, the latent-space visualizations transition from compact, well-separated clusters to diffuse and overlapping point clouds, and the descriptive summary translates the results into an interpretable explanation.

4 Representation Analysis for Real-world Discovery

We next apply LatentVerse to real biomedical embeddings to assess its value in better understanding and assessing real-world representations. We show that the diagnostic tests can reveal which modality carries the strongest signal for a given phenotype, whether that signal is shared across modalities or confined to one of them, how concentrated or diffuse it is within latent space, and how robust different embeddings are under analysis. These questions are highly relevant in biomedical machine learning, where pretrained representations are increasingly reused for downstream studies but are often treated as opaque features. To this end, we analyze pretrained embeddings from UK Biobank [5], derived from electrocardiogram (ECG) signals and cardiac magnetic resonance imaging (cMRI) using pretrained encoder models. Figure 4 summarizes the resulting analyses.

Figure 4 illustrates several examples of this analysis. Figure 4.a shows differences in predictive accessibility across modalities. For the PQ interval, ECG representations achieve substantially higher predictive performance than cMRI representations, with strong R^2 values even for simple probes, indicating that this signal is directly accessible from ECG latent space. In contrast, cMRI representations are less predictive for the same target, suggesting that the relevant information is either weaker or encoded less directly.

Figure 4.b provides insight into how information is distributed in latent dimensions

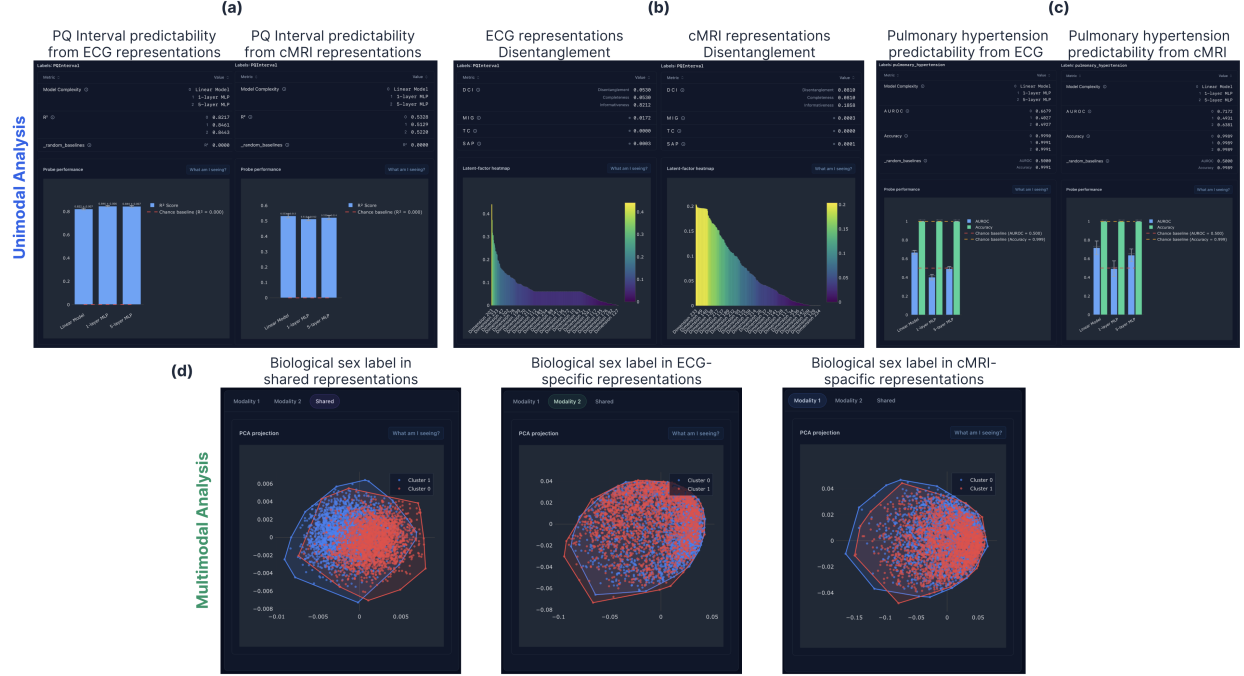


Figure 4: **LatentVerse analysis of real biomedical embeddings from UK Biobank.** (a) PQ-interval predictability from ECG and cMRI representations. (b) PQ-interval disentanglement in ECG and cMRI representations. (c) Pulmonary-hypertension predictability from ECG and cMRI representations. (d) Multimodal analysis of biological sex across shared, ECG-specific, and cMRI-specific latent spaces, showing stronger separation in the shared representation. These examples illustrate how LatentVerse localizes clinically relevant information across modalities and latent subspaces.

through disentanglement analysis. For the same PQ-interval target, the ECG representation exhibits stronger evidence of concentrated, target-related structure, whereas the cMRI representation shows weaker disentanglement and a higher distribution of information across latent dimensions. The corresponding heatmaps make this distinction visually apparent by showing whether label-related signal is localized to a smaller subset of latent variables or spread more broadly across the embedding.

Figure 4.c shows clinical predictability of different representations for pulmonary hypertension diagnosis. Both ECG and cMRI representations can be assessed through probing, but the reports also show why a single metric is insufficient: because the label is highly imbalanced, accuracy remains artificially high across probes, while AUROC is more informative about true predictive utility. In this example, cMRI representations appear to provide stronger discriminative signal than ECG representations, illustrating how the platform helps users identify which modality carries more clinically useful information.

The multimodal analysis further illustrates how LatentVerse can support discovery by localizing where specific biological information is encoded. Figure 4.d shows the distribution of biological sex labels across the shared, ECG-specific, and cMRI-specific representation components. In this case, the clearest separation appears in the shared latent space, while the modality-specific components show substantially weaker structure. This is consistent with the

expectation that biological sex is reflected across both ECG and cMRI and therefore should be captured primarily in information shared between the two modalities. More broadly, this example shows how LatentVerse can help distinguish whether a clinically relevant variable is jointly encoded across modalities or driven mainly by one modality alone.

5 User Study

To assess the practical utility of LatentVerse, we conducted a user study with researchers from diverse biomedical application domains who were already familiar with machine learning workflows. Rather than evaluating the system on a single task or dataset, the study was designed to understand how LatentVerse could support representation analysis in participants’ own areas of work. Participants used the platform to examine embeddings drawn from a range of biomedical applications and then provided feedback on how its metrics, visual reports, and comparative views could support their existing analytical practice. The study therefore focused on the perceived usefulness, interpretability, and workflow fit of LatentVerse for real biomedical representation analysis, rather than only on generic usability. The full study protocol and survey instrument are provided in Appendix A.7.

5.1 Results

Three researchers took part: two working in genomics and proteomics and one in clinical machine learning, all of whom currently evaluate embeddings using Python notebooks and custom visualisation scripts. Participants used the platform as deployed in July 2026. Given this sample size, we report their responses as formative design feedback rather than as a quantitative evaluation.

Participants responded positively to the tool’s usefulness and readability. They valued having standard metrics immediately available for quick iteration, the worked examples and contextual explanations, and the reporting of job progress; one noted that the evaluation pipeline surfaced metrics they had not previously encountered, and both participants who engaged with the rating scale expressed confidence in the conclusions they drew from it.

Their reservations was mainly on verifiability rather than usability. Participants could not trace a reported finding back to the individual samples behind it, and one observed that the reports “take some trust that the tool correctly generated the answers.” A second was unsure whether the groups shown in the clusterability view were unsupervised clusters or label classes. Participants also asked for pre-specified subgroup analyses and for a scripted path to reproduce an analysis inside a version-controlled repository. Per-participant responses are provided in Appendix A.7. These observations shaped the platform’s subsequent development. The current deployment reports how each metric was computed through a provenance panel, lets a reported finding be traced back to the individual samples behind it through per-sample export, distinguishes unsupervised clusters from label classes in the clusterability view, and adds pre-specified subgroup comparison. The command-line interface provides the scripted, version-controlled path for reproducing an analysis that participants requested.

6 Discussion

We introduced LatentVerse, a framework for unimodal and multimodal representation analysis that unifies diverse metrics for evaluating representation quality within a single report-driven workflow. By combining quantitative diagnostics, visual summaries, comparative analysis, and descriptive interpretation, LatentVerse supports systematic analysis of latent spaces for users with different levels of technical expertise. This makes representation analysis useful not only for model evaluation and debugging, but also for discovery-oriented analysis. Our controlled simulations show that LatentVerse helps recover the structural properties of synthetic representations, and our biomedical analyses showed how the same diagnostics characterize what real-world embeddings encode and how the information is embedded. We also conducted a user study to assess the utility of LatentVerse for embedding evaluation in different application domains, and are using the feedback from this study to guide the next iteration of the platform.

The most distinctive contribution of LatentVerse is its support for multimodal representation analysis. As foundation models increasingly combine multiple data modalities, understanding how information is distributed across modalities becomes essential for meaningful validation and trust. LatentVerse addresses this challenge by decomposing fused embeddings into shared and modality-specific components, making it possible to analyze where predictive or biologically relevant signal resides rather than treating the multimodal representation as a single opaque vector. We believe this capability is especially important because it addresses a growing need that is not well served by existing tools and is likely to remain relevant across models, modalities, and application domains.

6.1 Limitations and future directions

LatentVerse is currently a research prototype and although the platform supports interactive evaluation of latent representations through a web application and command-line interface, further engineering will be needed to support broader deployment across larger user bases, more demanding workloads, and very large embedding datasets. In particular, scaling the platform to support more concurrent users, larger multimodal representations, and more computationally intensive analyses will require additional infrastructure, job scheduling, and resource management.

In terms of utility of LatentVerse in different application domains, our user study currently provides an initial assessment of the platform’s practical value across researchers familiar with machine learning workflows, but it does not yet establish how well the tool transfers across the full range of scientific and application domains in which latent representations are used. Extending this evaluation to additional domains, user profiles, and analytical settings will be a future direction for our work that will also help further develop the application.

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usability, and evaluation of LatentVerse.

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Author Contributions

M.A. developed the LatentVerse platform, including the full-stack web application and backend, reworked the implementation of the evaluation tests and integrated them into the platform. M.A. also handled data curation, helped with the visualizations, validated the test implementations, ran the experiments, and led the writing of the paper as well as the conceptualization and methodology. S.F. Helped with running the real-world discovery analysis. J.K. helped with the Google Cloud Platform deployment. S.T. supervised the research, shaped and guided the methodology for the experiments and contributed to writing the paper. M.M. supervised the research and wrote the paper.

Declaration of Interests

The authors declare no conflict of interest.

Resource Availability

The LatentVerse evaluation library is openly available at <https://github.com/broadinstitute/ml4h-latentverse>. The web application is available at <https://broad.io/latentverse>.

UK Biobank data (Section 4) are available to approved researchers through the UK Biobank (<https://www.ukbiobank.ac.uk>) under application number 7089.

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A Supplemental Information

A.1 System architecture and implementation

LatentVerse is organized as a three-tier web application consisting of a React-based frontend, a Python FastAPI backend, and a standalone machine learning library. The backend exposes RESTful endpoints that handle file uploads, input validation, test execution, and result serialization. The primary endpoint runs the full evaluation pipeline: it locates uploaded files by request identifier, loads and preprocesses the CSV data, merges representation and label files on user-specified ID columns, and dispatches the selected benchmark test.

File ingestion uses a three-tiered strategy keyed on file size. For files larger than 50 MB, LatentVerse switches to Polars with eight parallel threads, achieving 10–100× faster loading compared to pandas for large tabular files. Files between 10 and 50 MB are handled by PyArrow with a 1 MB block size and self-destructing buffers to conserve memory. Smaller files fall back to the pandas C engine.

When multiple label columns are selected for a single test run, the backend evaluates all columns concurrently using `asyncio.gather` combined with a thread pool executor, so wall-clock time scales with the most expensive label rather than the sum of all labels. To keep this concurrency from fighting with itself, BLAS thread counts are capped at one when more than one label is being evaluated in parallel, which avoids the oversubscription pattern where

NumPy and scikit-learn each spin up a full thread pool inside every asyncio task. Concurrency is also bounded by a configurable semaphore (`MAX_PARALLEL_LABEL_TESTS`), and per-worker thread budgets are capped to prevent BLAS oversubscription. Disentanglement statistics (entropy, mutual-information gaps) are accelerated with Numba just-in-time compilation, giving a 10–20× speedup over the pure-Python path on large representations.

Long-running comparisons run asynchronously. The frontend submits the job, immediately receives a job ID, and polls a lightweight status endpoint that reports both a percentage and a named phase (`queued`, `loading_files`, `decoupling`, `running_tests`, `generating_plots`, `done`). Job state is tracked in an in-memory store so that a single instance owns each job for its lifetime, which is also why the deployment pins session affinity. Multimodal training is off-loaded to a Cloud Run Job via Pub/Sub, writing inputs and status to GCS and surfacing progress through the same job pipeline that unimodal tests use, so the frontend treats both cases identically.

The application is deployed on Google Cloud Run, configured with 8 vCPUs, 16 GiB of memory, autoscaling between zero and ten instances, and CPU always allocated (`-no/cpu-throttling`) so background jobs continue to make progress between polls. Uploaded files are persisted to a Google Cloud Storage bucket mounted with GCS-FUSE; completed files are reaped after 48 hours. For institutional deployments, the platform exposes an optional authenticated mode with per-user run history and Identity-Aware Proxy support, currently disabled in the public deployment. When a compatible GPU is present (NVIDIA L4 on the production Cloud Run service), MultiLoReFT training runs on-device via PyTorch’s CUDA backend; the service auto-detects hardware at startup and routes computation accordingly.

A.2 Web application interface

The frontend is a single-page application organized around the three-step workflow described in Section 2.2. In the first step, the user selects an evaluation test; for several tests, contextual messages explain the label requirements before any files are uploaded. A guidebook is reachable from the test selector, each comparison card, and the results panel, providing metric descriptions, score ranges, interpretation guidance, and common pitfalls. Dark and light themes are supported, a two-stage reset confirmation prevents accidental loss of a configured comparison, and an example mode loads pre-computed results for all five unimodal tests and the multimodal workflow, annotated with interpretive bullet points.

In the second step, the user configures one or more comparison cards. Each card accepts an experiment name, a representation file in CSV or TSV format, and an optional labels file or a small set of comma-separated numeric labels. After uploading, the application parses the first 100 rows client-side using PapaParse to auto-detect column headers, identify headerless files, and flag plausible label columns. A searchable dropdown lets the user select ID columns for matching representations and labels; when files are guaranteed to be in the same row order, an injected virtual column can be used for index-based alignment. An “advanced” drawer exposes run options that affect reproducibility and runtime: a random seed, an in-place standardization toggle, a row subsample cap, and (for clustering-based tests) an override for the number of clusters. When labels are joined on an ID column, the backend tracks how many representation rows fail to match and surfaces a warning whenever overlap falls below a configurable threshold. File transfers use the TUS resumable upload protocol; each

file is fingerprinted by name, size, and last-modified timestamp and is not re-uploaded if an identical fingerprint is cached and confirmed via a `/validate-files` check.

In the third step, results are returned as a JSON payload rendered into four components: an interactive Plotly.js chart whose form depends on the test (PCA scatter with cluster hulls for clusterability, mutual-information heatmaps and sorted bar charts for disentanglement, dimension-removal curves for expressiveness, grouped bar charts with cross-validation error bars for probing, and noise-degradation line charts with a 90%-of-baseline threshold for robustness); a sortable metrics table; export buttons for raw JSON and formatted CSV; and an optional AI-generated summary produced by querying the Google Gemini 2.5 Flash via the Vertex AI “generateContent” endpoint (service-account / ADC auth). The Gemini prompt is built server-side from a fixed template that includes the experiment name, metric definitions with expected ranges, test-specific interpretation guidance, and the computed values; the model is instructed to produce a 3–5 sentence technically grounded paragraph that references the experiment by name and avoids naive good/bad characterizations. The proxy fronting the Gemini API enforces this system instruction server-side, strips user-supplied prompt overrides, and applies a token-bucket rate limit so the summary feature cannot be repurposed as an open LLM endpoint.

A.3 Representation evaluation tests

A.3.1 Clusterability

LatentVerse assesses clusterability using both intrinsic and label-aware metrics. The Silhouette Score [23] and Davies–Bouldin Index [10] quantify cluster cohesion and separation directly from the geometry of the representation; for $N > 5,000$, Silhouette is estimated on a 20% random subsample (minimum 1,000 samples) for tractability. When ground-truth labels are available, NMI [26] measures alignment between discovered clusters and known classes. We additionally report Cluster Learnability [18], the balanced accuracy of a logistic-regression probe trained on a stratified 80/20 split, which measures how easily class labels can be recovered. In label-free settings, the analysis relies on the intrinsic metrics alone.

A.3.2 Disentanglement

LatentVerse reports the DCI framework [12] (Disentanglement, Completeness, Informativeness), the Mutual Information Gap (MIG) [8], Separated Attribute Predictability (SAP) [16], and Total Correlation (TC) [32]. Formally, for generative factor y_j ,

$$\text{MIG}_j = \frac{I(z; y_j)^{(1)} - I(z; y_j)^{(2)}}{H(y_j)}, \quad (1)$$

where $I(z; y_j)^{(1)}$ and $I(z; y_j)^{(2)}$ are the largest and second-largest mutual-information values between any single latent dimension and y_j , and $H(y_j)$ is the entropy of the factor. The implementation supports $N \geq 1$ simultaneous generative factors. For $N > 1$, Disentanglement and Completeness follow the standard DCI weighting [12] rather than a plain average: Disentanglement is the per-dimension score (one minus the normalized entropy of a dimension’s relative importance across factors), averaged over dimensions and weighted by each dimension’s

total importance; Completeness is the per-factor score (one minus the normalized entropy of a factor’s relative importance across dimensions), averaged over factors and weighted by each factor’s total informativeness. Informativeness, MIG, and SAP are averaged over factors with equal weights, and Total Correlation is computed once for the full representation. Mutual information is estimated via k -nearest-neighbour regression and classification ($k = 3$) using `mutual_info_regression` and `mutual_info_classif` from scikit-learn [20]. TC is approximated via the Gaussian proxy $\frac{1}{2} \log\left(\prod_i \sigma_i^2 / \det(\Sigma)\right)$, which is exact under Gaussian marginals and avoids the cost of GMM-based estimation.

A.3.3 Expressiveness

LatentVerse progressively removes the highest-variance dimensions and measures the resulting degradation in predictive performance. The removal schedule sweeps $\{0\%, 10\%, 20\%, 30\%, 40\%, 50\%\}$ of dimensions by default. Compactness measures the average performance degradation:

$$\text{Compactness} = 1 - \frac{1}{|S|} \sum_{s \in S} \frac{\text{perf}(s)}{\text{perf}(0)}, \quad (2)$$

where $S = \{0\%, 10\%, 20\%, 30\%, 40\%, 50\%\}$, $\text{perf}(s)$ is the downstream performance when $s\%$ of dimensions are removed, and $\text{perf}(0)$ is the full-representation baseline; the score is clamped to $[0, 1]$. Intrinsic Dimension estimates how many dominant dimensions can be removed before performance deteriorates substantially.

A.3.4 Downstream predictability

LatentVerse uses supervised probes of increasing complexity, from a linear probe to a five-layer MLP. Each probe is evaluated with three-fold cross-validation and the mean and standard deviation are reported. Task type (binary classification, multi-class classification, or regression) is detected automatically from the label column, and the test reports Accuracy, AUROC, macro-F1, and R^2 as appropriate.¹

A.3.5 Robustness

LatentVerse injects Gaussian noise of increasing magnitude into the latent space at levels $\sigma \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. In clustering mode, robustness is reported through Silhouette Score, Davies–Bouldin Index, and (when labels are available) NMI. In probing mode, it is evaluated using Accuracy, AUROC, macro-F1, or R^2 , depending on the task. For visualization, a representation is considered robust at a given noise level if its performance remains at or above 90% of the clean-data baseline.

A.4 MultiLoReFT configuration

Table 1 summarizes all parameters used for the MultiLoReFT framework to decouple the multimodal information for analysis.

¹Exposed as `run_probing` in the underlying LatentVerse library; we use the more descriptive name throughout the paper.

Table 1: MultiLoReFT configuration used in our experiments.

Hyperparameter	Value
Shared rank r_s	4
Modality-specific rank r_m	4
Learning rate	1×10^{-3}
Effective batch size	1024
Singular-value pruning threshold	0.1
Restarts (best-of- N)	1
Max epochs / training timeout	800 / 1200 s

A.5 Simulated datasets for controlled validation

A.5.1 Unimodal datasets

Each dataset contains 5,000 samples, 64 real-valued features, and discrete integer labels, grouped into four families targeting clusterability, predictability, disentanglement, and expressiveness, with three difficulty levels each. The clusterability datasets use 5 labels (0–4) and vary how strongly labels correspond to separable clusters. The predictability datasets use 10 labels (0–9) and vary the complexity of the feature-to-label mapping from linearly separable to nonlinear and interaction-driven. The disentanglement datasets use 10 labels and vary whether label-related information is encoded in a single axis-aligned factor, spread across multiple observed dimensions, or mixed across latent factors. The expressiveness datasets derive 10 labels from a continuous latent score and vary how compactly predictive information is distributed across the 64 dimensions.

A.5.2 Multimodal dataset

The simulation generates two modalities from shared and modality-specific latent variables sampled from non-Gaussian distributions. In total $n_{\text{hidden}} = 2 + 2 + 2 = 6$ hidden variables are defined: two shared, two private to modality 1, and two private to modality 2. The shared variables $z_s \in \mathbb{R}^2$ are sampled as $z_s \sim \text{Binomial}(1, 0.5) + \mathcal{N}(0, 0.01I)$. Modality-specific factors are drawn from distinct non-Gaussian distributions: $z_{m1} \sim \text{Weibull}(1.5) \times 0.3$ and $z_{m2} \sim \text{Beta}(3, 2)$. Each point is annotated with four categorical labels:

$$y_s \in \{0, 1, 2, 3\}, y_{m1} \in \{0, 1, 2, 3\}, y_{m2} \in \{0, 1, 2, 3\}, y_{\text{joint}} \in \{0, \dots, 7\}.$$

y_s is determined by the unique binary configurations of the un-noised shared factors. To obtain categorical modality-specific labels, the private continuous factors are binarized dimension-wise using the empirical median threshold, yielding a 2-bit codeword per modality; y_{m1} and y_{m2} are assigned as the index of the corresponding unique codeword. The joint label y_{joint} is defined by the triple (y_s, y_{m1}, y_{m2}) : each unique triple is mapped to a joint class ID, deterministically bucketed (via a seed-controlled permutation) into at most $K = 8$ joint classes. This ensures y_{joint} depends on all three other labels and is not predictable from either modality alone.

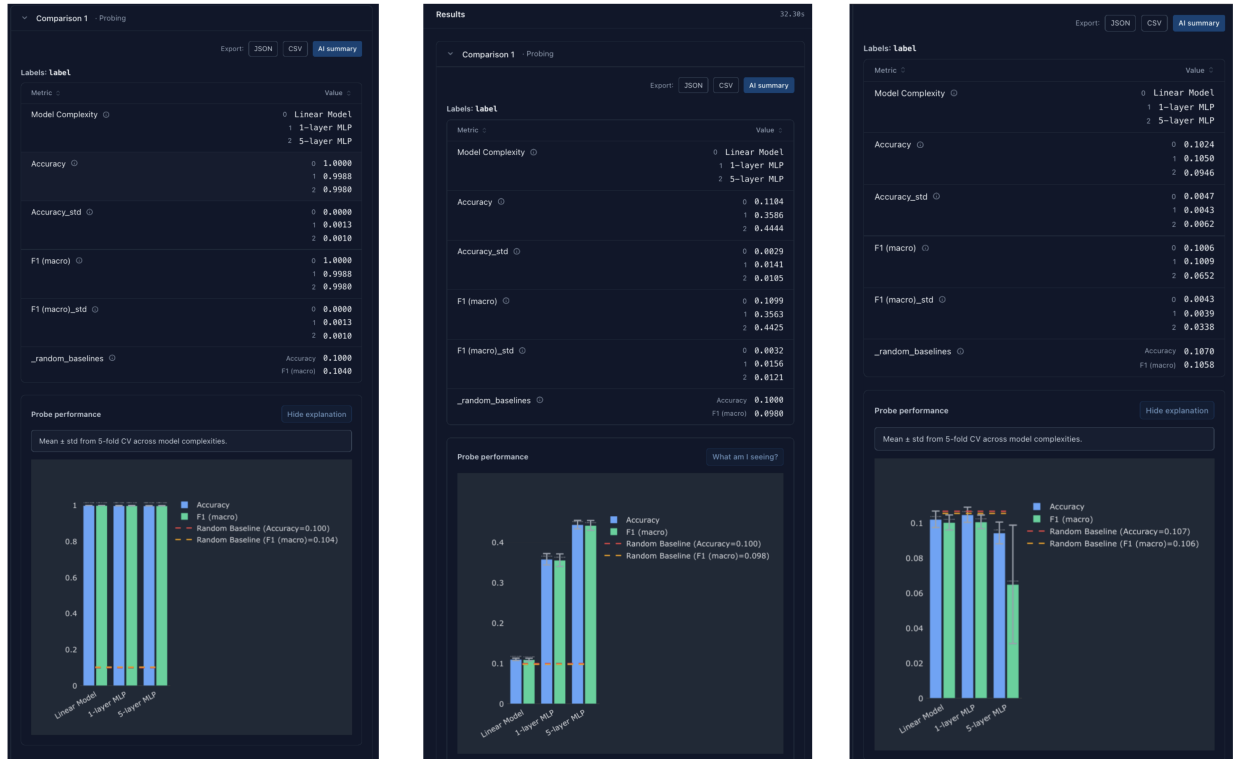


Figure 5: LatentVerse reports for probing-based predictability evaluation across representations with low, moderate, and high predictive accessibility. Each report combines a numerical summary of probe performance across model complexities with a visual comparison of accuracy and macro-F1 against random baselines.

A.6 Controlled offline validation supplement

Predictability. Across the predictability datasets, LatentVerse distinguishes directly accessible labels (linear probe accuracy = 1.0000) from labels that require more expressive decoders (accuracy rising from 0.1104 for the linear probe to 0.4392 for the 5-layer MLP) and from uninformative representations where all probes remain near random, with a drop in macro-F1 for the most complex probe indicating instability or overfitting (Figure 5).

Disentanglement. Across the disentanglement datasets, the easiest setting shows higher DCI, MIG, and SAP with lower Total Correlation, consistent with signal concentrated in a small number of dimensions; intermediate and hard settings show weakening scores and increasing cross-dimension dependence, reflecting progressively more entangled representations (Figure 6).

Expressiveness. In the compact setting, performance drops sharply under ablation (0.6248 baseline to 0.1901 after removing 50% of dominant dimensions; compactness 0.4672, intrinsic dimension 6). In the semi-compact, correlated setting, performance is largely insensitive to removal (0.7149 to 0.6637; compactness 0.0335, intrinsic dimension 32). In the not-compact,

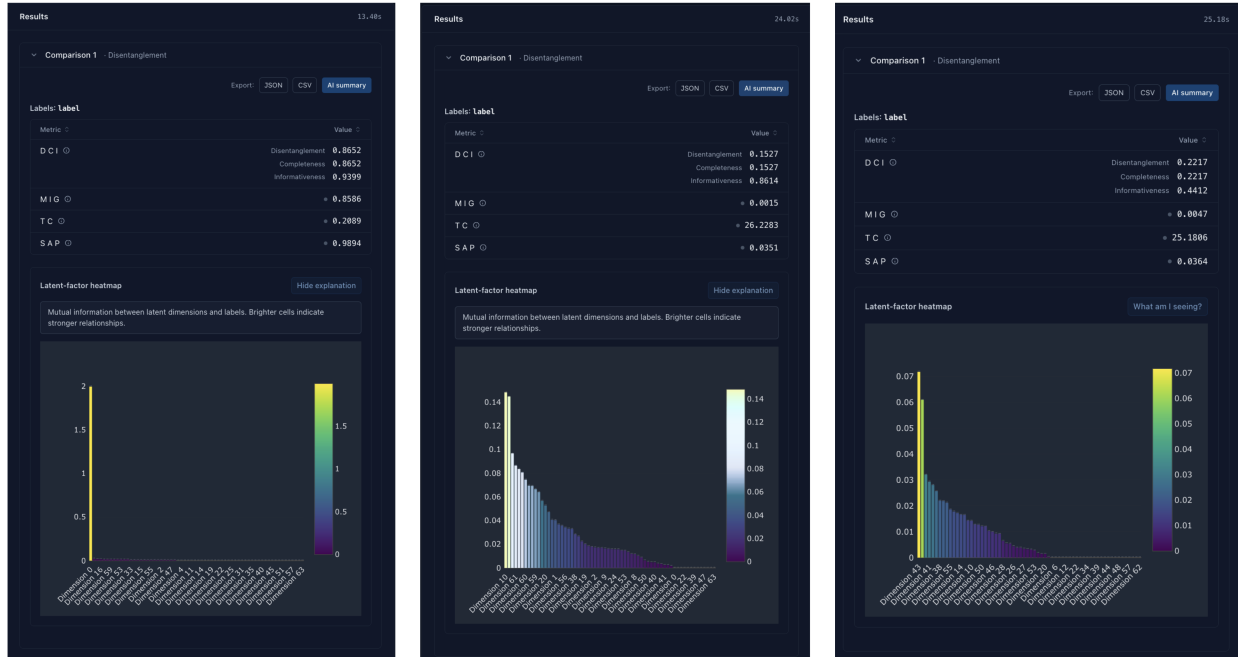


Figure 6: LatentVerse reports for disentanglement evaluation across representations with low, moderate, and high factor entanglement, combining DCI, MIG, SAP, and Total Correlation with visual summaries of how label-related information is distributed across latent dimensions.

sparse setting, performance again drops sharply (0.6211 to 0.1833; compactness 0.3491, intrinsic dimension 13) (Figure 7).

A.7 User-study protocol and survey instrument

To evaluate the usability, interpretability, and perceived utility of LatentVerse, we designed a short post-use feedback survey, administered after participants explored the tool using their own latent representations or embeddings. The survey was designed to understand (i) what users typically try to evaluate in embeddings, (ii) whether LatentVerse supports those goals, (iii) whether the tool helps users interpret and compare representations, and (iv) how it compares with users’ existing workflows.

The survey introduced LatentVerse as a web application for helping developers and researchers analyze and evaluate latent representations through visualizations and quantitative metrics presented in a structured report. Participants were informed that the tool supports comparison, interpretation, and validation of representation properties such as clustering structure, robustness, predictability, expressiveness, and, when relevant, multimodal structure. Participants were also told that responses would be analyzed in aggregate and that anonymized quotes could be used in a research publication.

Survey introduction and consent

Participants first received brief instructions for using the tool. They were told that LatentVerse can be accessed through the web interface and that, in general, the tool requires an embedding

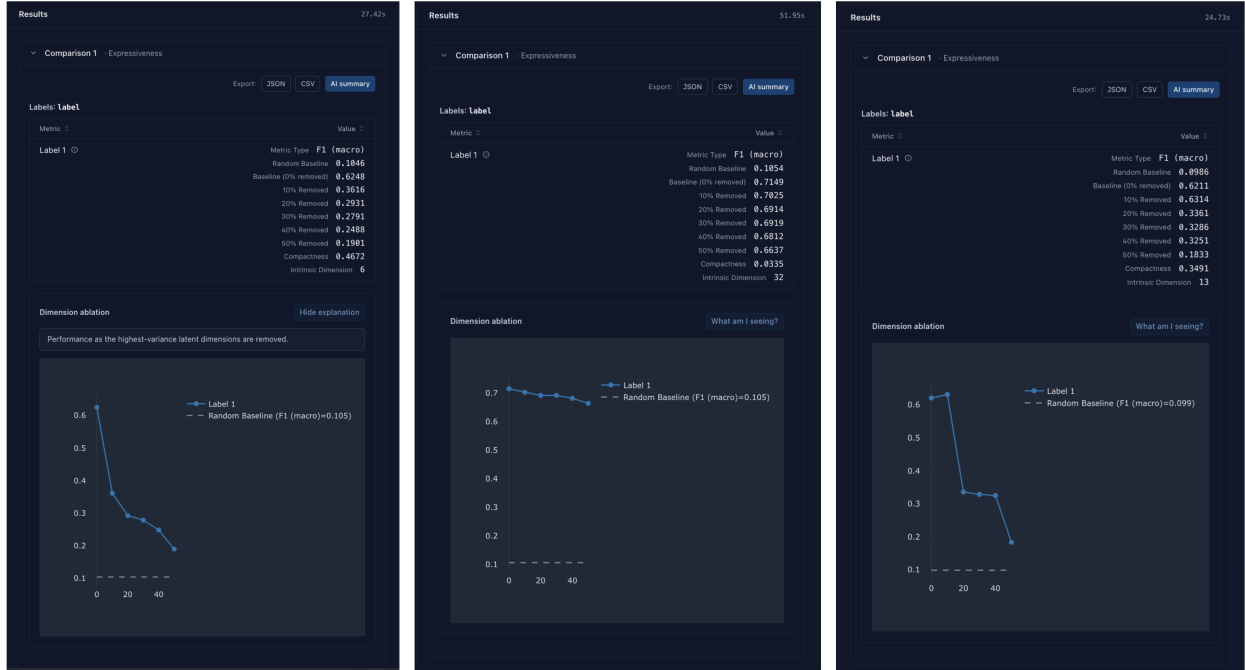


Figure 7: LatentVerse reports for expressiveness evaluation across three synthetic datasets (compact, correlated-redundant, and sparse-informative), combining baseline performance, compactness, and intrinsic dimension with an ablation curve showing how predictive performance changes as high-variance dimensions are progressively removed.

file in CSV format, where each row corresponds to a sample, the first column contains a sample identifier, and the remaining columns contain embedding dimensions. Participants were also informed that they could optionally upload a label file with one or more labels per sample, such as class, cohort, site, outcome, or subgroup. These labels support analyses such as separability, subgroup comparisons, and predictability, although some analyses can be run without labels.

Before completing the survey, participants were asked to provide consent for their anonymized responses to be used for research and publication purposes.

1. **Consent.** I consent to my responses being used, in anonymized form, for research/publication purposes.
 - Yes, I consent.
2. **Optional follow-up.** You may contact me for a short follow-up discussion.
 - Yes, you may contact me.
3. **Email address.** Participants who agreed to follow-up contact were asked to provide an email address.
4. **Recommendation of other users.** Participants were also asked whether they could recommend someone else who may be willing to test LatentVerse.

- Yes, I can recommend someone else.
5. **Recommended contact.** Participants who selected this option were asked to provide the recommended person's email address.

About the participant and their workflow

The first section of the survey collected information about participants' application domains, machine learning workflows, embedding types, and current evaluation practices.

1. What is your primary application domain?

- Clinical / healthcare
- Biology / genomics / proteomics
- Public health / epidemiology
- Neuroscience
- General ML / non-biomedical
- Other

2. What type(s) of ML are you working with most often?

- Supervised learning
- Self-supervised / contrastive learning
- Representation learning / embeddings as a primary output
- Generative models
- Multimodal models, e.g., text+image or EHR+labs
- Time series models
- Graph models
- LLMs / transformers
- Other

3. What do your embeddings most often represent?

- Patients / individuals
- Visits / time windows
- Images / patches
- Text documents / notes
- Cells / genes / proteins
- Samples / specimens
- Other

4. When in your workflow do you usually evaluate embeddings?

- Model development / debugging
- While choosing between models or checkpoints
- Before downstream training
- During scientific analysis / hypothesis generation
- For reporting in a paper / benchmarking
- Teaching / communication
- Rarely / never
- Other

5. Why do you evaluate embeddings? What decision does it support?

Free-text response, 1–3 sentences. Participants were given examples such as selecting the best model, understanding what is encoded, checking robustness, comparing checkpoints, verifying biological signal, or detecting subgroup effects.

6. What evaluations do you typically perform on embeddings?

- Downstream performance, e.g., linear probe, classifier, or regressor
- Visualization, e.g., UMAP, t-SNE, PCA, and manual inspection
- Clustering and cluster quality metrics
- Nearest-neighbor inspection / retrieval sanity checks
- Subgroup / cohort comparisons
- Outlier / anomaly detection
- Robustness checks, e.g., augmentation, noise, or shift
- Interpretability / disentanglement analyses
- Bias / fairness checks across subgroups
- Calibration / uncertainty of embedding-based decisions
- Other

7. What tools do you currently use for these evaluations?

- Python notebooks, e.g., NumPy, PyTorch, or scikit-learn
- TensorBoard projector or similar embedding viewers
- Custom visualization scripts
- Commercial / GUI analytics tools
- I do not have a standard approach
- Other

8. What usually makes embedding evaluation hard in your setting?

Participants were asked to select up to two options.

- Too many ad-hoc scripts / poor reproducibility
- Hard to compare models consistently
- Hard to interpret what metrics mean
- Hard to connect results back to specific samples
- Takes time to implement validation each time
- Unclear what to check / lack of guidance
- Hard to reason about multimodal embeddings
- Other

9. Can you briefly describe the embeddings you tested today?

Free-text response. Participants were asked to describe what the embeddings represent, the approximate dataset size, and whether the setting was unimodal or multimodal.

Session-specific feedback on LatentVerse

The second section of the survey asked participants about the analyses they used during their LatentVerse session, the conclusions they reached, and how the tool compared with their usual workflow.

10. Which analyses or views did you use today?

- Clusterability
- Disentanglement
- Expressiveness
- Downstream predictability / probing
- Robustness
- Multimodal shared vs. modality-specific analysis
- Visualization of latent space / projections
- LLM-generated summary / interpretation
- Other

11. How long did it take you to reach a useful conclusion with LatentVerse?

- < 5 minutes
- 5–10 minutes
- 10–20 minutes
- 20 minutes
- I did not reach a useful conclusion

12. Rate the following statements.

Participants rated each statement on a 1–5 Likert scale, where 1 indicated “strongly disagree” and 5 indicated “strongly agree.”

- LatentVerse helped me answer the evaluation question I cared about.
- I could use LatentVerse to compare representations systematically.
- I understood what the reported metrics and views were showing.
- The tool provided enough evidence to support the conclusions I reached.
- I trust the conclusions I reached using LatentVerse.
- LatentVerse would be useful in my regular workflow.

13. Compared with your usual workflow, please rate the following.

Participants rated each statement on a 1–5 Likert scale, where 1 indicated “strongly disagree” and 5 indicated “strongly agree.” For the multimodal-analysis item, participants were instructed to select neutral if the question was not applicable.

- LatentVerse was faster for this kind of analysis.
- LatentVerse made it easier to compare representations.
- LatentVerse made the results easier to interpret.
- LatentVerse provided clearer evidence than my usual approach.
- LatentVerse helped me reach a more confident conclusion.
- LatentVerse would be especially useful for multimodal embedding analysis.

14. What conclusion or decision were you able to reach using LatentVerse?

Free-text response, 1–3 sentences. Participants were given examples such as determining that one model appeared more robust, that a shared representation space was more informative than a modality-specific one, or that an embedding had weak cluster structure.

15. Where did you feel least confident or most confused?

Free-text response.

16. What did you find most useful about LatentVerse?

Free-text response.

17. What else would you have liked to do with LatentVerse?

Free-text response. Participants were prompted to describe what additional information, metric, visualization, or interaction would have made the evaluation more convincing.

18. What is the most important improvement you would like to see?

Free-text response.

19. Where do you see LatentVerse fitting into your lab, team, or workflow?

Free-text response.

A.7.1 Survey Results

Table 2: Per-participant responses (1–5). P1 returned the neutral midpoint on all eight statements and reported evaluating embeddings “rarely or never.”

Statement	P1	P2	P3
Helps achieve my evaluation goal	3	4	4
I could complete my evaluation	3	5	3
I trust the conclusion I reached	3	5	5
Overall usefulness in my workflow	3	4	4
I understood what each view/metric showed	3	5	4
The UI provided enough context	3	5	4
I could see the evidence behind conclusions	3	3	5
I could trace findings back to samples	3	3	3
Ease of use (UI/UX)	4	4	4